Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

Alex Vergara

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1. **For each of the following matrices:**

   - **Obtain a row-equivalent matrix in echelon form using elementary row operations.**
   - **Determine its rank.**
   - **Obtain a row-equivalent matrix in reduced echelon form using elementary row operations.**

a) $A = \begin{bmatrix} 2 & 4 & 0 & 3 \\ 0 & 0 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ -1 & 4 & 0 & 4 \end{bmatrix}$

i. **Row Echelon:**

$$\text{A} \xrightarrow[R_2 \leftrightarrow R_4]{R_1 \leftrightarrow R_3}{}_{t_1} \begin{bmatrix} 1 & 2 & 0 & 2 \\ -1 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 \\ 0 & 0 & 2 & 3 \end{bmatrix} \xrightarrow[t_2]{R_2 \leftarrow R_2 + R_1} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 6 & 0 & 6 \\ 2 & 4 & 0 & 3 \\ 0 & 0 & 2 & 3 \end{bmatrix}$$

$$\xrightarrow[t_3]{R_4 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 6 & 0 & 6 \\ 0 & 0 & 2 & 3 \\ 2 & 4 & 0 & 3 \end{bmatrix} \xrightarrow[t_4]{R_4 \leftarrow R_4 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 6 & 0 & 6 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

ii. **Rank:** $\operatorname{rank}(A) = 4$.

iii. **Reduced Echelon form:**

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 6 & 0 & 6 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix} \xrightarrow[R_1 \leftarrow 2R_2 - R_1]{R_2 \leftarrow \frac{1}{6}R_2}{}_{t_5} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix} \xrightarrow[R_2 \leftarrow R_4 - R_2]{R_3 \leftarrow 3R_4 + R_3 \, R_3 \leftarrow \frac{1}{2}R_3}{}_{t_6} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\xrightarrow[t_7]{R_4 \leftarrow -1R_4} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b)
$$B = \begin{bmatrix} 0 & 0 & 1 \\ 4 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

c)
$$C = \begin{bmatrix} 0 & 0 & 3 & 3i & -15 \\ i & 3 & 2 & 0 & 3 \\ 2 & -6i & -5i & 1 & 2 \end{bmatrix}$$

d)
$$D = \begin{bmatrix} 2 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 \\ 4 & -4 & 4 & 4 \\ 0 & 2 & 0 & 0 \end{bmatrix}$$

e)
$$E = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

f)
$$F = \begin{bmatrix} 3 & 0 & 0 & -21 & 24 \\ 4 & 1 & 0 & -25 & 34 \\ -2 & 3 & 0 & 22 & -2 \end{bmatrix}$$

18. **Consider the following matrices.**

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -1 & 1 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -2 & -4 & -6 & -2 \\ -2 & -1 & -1 & 0 \end{bmatrix}$$

a) Determine elementary matrices H, J, K such that $HA = B$, $JA = C$ and $KA = D$.

Let $T_{jk}(c)$ be I_3 with entry (j,k) replaced by c, for $j \neq k$. Left multiplication by $T_{jk}(c)$ performs $F_j \leftarrow F_j + cF_k$. Let S_{23} interchange rows $2, 3$ and let $D_j(c)$ multiply row j by c. These are elementary matrices when $c \neq 0$ in $D_j(c)$. Here

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}.$$

B is obtained by interchanging rows $1, 2$; C by $F_2 \leftarrow F_2 + F_3$; D by $F_2 \leftarrow -2F_2$. Therefore

$$H = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad J = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Direct multiplication gives $HA = B$, $JA = C$, $KA = D$.

b) Determine, if they exist, elementary matrices $E_p, \ldots, E_1$ such that

i)

$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

Apply $F_2 \leftarrow F_2 - F_1$, $F_3 \leftarrow F_3 + 2F_1$ and $F_2 \leftrightarrow F_3$. Thus

$$S_{23}T_{31}(2)T_{21}(-1)A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 2 & 2 & 0 \end{bmatrix}.$$

Take $E_1 = T_{21}(-1)$, $E_2 = T_{31}(2)$, $E_3 = S_{23}$.

ii)

$$E_p \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

Continue from (i) with $F_3 \leftarrow F_3 + 2F_2$, $F_2 \leftarrow -3F_2$, $F_3 \leftarrow \frac{7}{4}F_3$. Hence take

$$E_1 = T_{21}(-1), \quad E_2 = T_{31}(2), \quad E_3 = S_{23},$$
$$E_4 = T_{32}(2), \quad E_5 = D_2(-3), \quad E_6 = D_3(7/4).$$

Indeed,

$$E_6 \cdots E_1 A = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 3 & -3 & -6 \\ 0 & 0 & 7 & 7 \end{bmatrix}.$$

The handwritten resolution prints 0 in position $(1, 4)$ of its final matrix. The target in the exercise sheet and the stated operations give 1 there.

c) Determine a matrix Q such that $QA = F$, where

$$F = \begin{bmatrix} -2 & 4 & 2 & -2 \\ 0 & 2 & 2 & 0 \\ 0 & 3 & 9 & 6 \end{bmatrix}.$$

Apply to $[A \mid I_3]$ the operations

$$F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 + 2F_1, \quad F_1 \leftarrow F_1 - F_2,$$
$$F_3 \leftarrow F_3 + F_2, \quad F_1 \leftarrow -2F_1, \quad F_3 \leftarrow 3F_3.$$

The result is

$$\left[\begin{array}{cccc|ccc} -2 & 4 & 2 & -2 & -4 & 2 & 0 \\ 0 & 2 & 2 & 0 & -1 & 1 & 0 \\ 0 & 3 & 9 & 6 & 3 & 3 & 3 \end{array}\right].$$

Consequently $Q = \begin{bmatrix} -4 & 2 & 0 \\ -1 & 1 & 0 \\ 3 & 3 & 3 \end{bmatrix}$ satisfies $QA = F$.

19. **Discuss the rank of each of the following matrices in terms of the real parameters α and β occurring in it.**

a) $A = \begin{bmatrix} -5 & 1 & 8 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha - 1 \end{bmatrix}$

$A = \begin{bmatrix} -5 & 1 & 8 \\ 0 & \alpha & 0 \\ 0 & 0 & \alpha - 1 \end{bmatrix}$. If $\alpha \notin \{0, 1\}$, there are three pivots. For $\alpha = 0$ or 1, exactly two rows remain independent. Thus

$$\mathrm{r}(A) = \begin{cases} 2, & \alpha \in \{0, 1\}, \\ 3, & \alpha \notin \{0, 1\}. \end{cases}$$

b) $B = \begin{bmatrix} 4 & \alpha - 1 & 3 \\ 0 & \alpha^2 + 1 & \alpha - 1 \\ 0 & 0 & \alpha + 5 \end{bmatrix}$

$B = \begin{bmatrix} 4 & \alpha - 1 & 3 \\ 0 & \alpha^2 + 1 & \alpha - 1 \\ 0 & 0 & \alpha + 5 \end{bmatrix}$. Since $4 \neq 0$ and $\alpha^2 + 1 > 0$, the first two pivots always exist. Therefore $\mathrm{r}(B) = 2$ for $\alpha = -5$, and $\mathrm{r}(B) = 3$ otherwise.

c) $C = \begin{bmatrix} -1 & 1 & 3 \\ 1 & \alpha - 1 & -\alpha - 3 \\ -1 - \alpha & 2 - \alpha & 3\alpha + 5 \end{bmatrix}$

For $C = \begin{bmatrix} -1 & 1 & 3 \\ 1 & \alpha - 1 & -\alpha - 3 \\ -1 - \alpha & 2 - \alpha & 3\alpha + 5 \end{bmatrix}$, apply $F_2 \leftarrow F_2 + F_1$, $F_3 \leftarrow F_3 - (1+\alpha)F_1$, $F_3 \leftarrow F_3 + 2F_2$, $F_2 \leftrightarrow F_3$, $F_3 \leftarrow F_3 - \alpha F_2$. This gives

$$C \sim \begin{bmatrix} -1 & 1 & 3 \\ 0 & 1 & 2 - 2\alpha \\ 0 & 0 & \alpha(2\alpha - 3) \end{bmatrix}.$$

No division by a parameter was used. Hence $\mathrm{r}(C) = 2$ for $\alpha \in \{0, \frac{3}{2}\}$, and $\mathrm{r}(C) = 3$ otherwise.

d) $D = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 2\alpha + 6 & \beta^2 - 4 & \beta^2 + 9 \\ -1 & -2 & \beta^2 & \beta - 1 \end{bmatrix}$

For

$$D = \begin{bmatrix} 1 & 2 & -1 & 2 \\ 3 & 2\alpha + 6 & \beta^2 - 4 & \beta^2 + 9 \\ -1 & -2 & \beta^2 & \beta - 1 \end{bmatrix},$$

$F_2 \leftarrow F_2 - 3F_1$, $F_3 \leftarrow F_3 + F_1$ give

$$D \sim \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 2\alpha & \beta^2 - 1 & \beta^2 + 3 \\ 0 & 0 & \beta^2 - 1 & \beta + 1 \end{bmatrix}.$$

If $\beta = -1$, the third row is zero and the second is nonzero. If $\beta = 1$, the last two rows are independent exactly when $\alpha \neq 0$. If $\beta \notin \{-1, 1\}$ and $\alpha \neq 0$, there are three pivots. If $\beta \notin \{-1, 1\}$ and $\alpha = 0$, subtracting row 3 from row 2 produces the nonzero last entry $\beta^2 - \beta + 2 = (\beta - \frac{1}{2})^2 + \frac{7}{4}$; the last two rows are again independent. Therefore

$$\mathrm{r}(D) = \begin{cases} 2, & \beta = -1, \text{ or } (\beta = 1 \text{ and } \alpha = 0), \\ 3, & \text{otherwise.} \end{cases}$$

20. **Consider the matrices**

$$A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}.$$

a) Verify that B is the inverse of A.

Here $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$.

$$AB = \begin{bmatrix} 6-5 & -15+15 \\ 2-2 & -5+6 \end{bmatrix} = I_2, \quad BA = \begin{bmatrix} 6-5 & 10-10 \\ -3+3 & -5+6 \end{bmatrix} = I_2.$$

Thus $B = A^{-1}$.

b) Is the matrix $\begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$ the inverse of A? Justify your answer.

For $D = \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$, $AD = \begin{bmatrix} 1 & -5 \\ 0 & -1 \end{bmatrix} \neq I_2$, so $D \neq A^{-1}$.

c) Determine $(5A)^{-1}$ and $(A^T)^{-1}$.

$$(5A)^{-1} = \tfrac{1}{5}B = \begin{bmatrix} 2/5 & -1 \\ -1/5 & 3/5 \end{bmatrix}, \qquad (A^T)^{-1} = B^T = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}.$$

d) Compute C^3. Is C invertible? What is its inverse?

$$C^2 = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}, \quad C^3 = I_2.$$

Therefore $C^{-1} = C^2$, since $CC^2 = C^2C = I_2$.

e) Is AC invertible? Justify your answer and, if it is, determine $(AC)^{-1}$.

The product is invertible, with

$$(AC)^{-1} = C^{-1}A^{-1} = C^2B = \begin{bmatrix} -3 & 8 \\ -2 & 5 \end{bmatrix}.$$

Indeed $(AC)(C^{-1}A^{-1}) = I_2 = (C^{-1}A^{-1})(AC)$.

21. **Consider two diagonal matrices of order n:**

$$A = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 & 0 & \cdots & 0 \\ 0 & b_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{bmatrix}.$$

a) Compute AB.

If $A = \operatorname{diag}(a_1, \ldots, a_n)$ and $B = \operatorname{diag}(b_1, \ldots, b_n)$, then $(AB)_{ij} = \sum_k a_{ik} b_{kj}$ vanishes when $i \neq j$ and equals $a_i b_i$ when $i = j$. Thus $AB = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n)$.

b) What conditions must $a_1, a_2, \ldots, a_n$ satisfy for A to be invertible? When A is invertible, what is A^{-1}?

A is invertible exactly when every $a_j \neq 0$, and then

$$A^{-1} = \operatorname{diag}(a_1^{-1}, \ldots, a_n^{-1}).$$

Both products with A equal I_n. If $a_j = 0$, row j of AX is zero for every X, so AX cannot equal I_n.

22. **Let $A_1, A_2, \ldots, A_n$ be invertible square matrices of the same order. Express $(A_1 A_2 \cdots A_n)^{-1}$ in terms of $A_1^{-1}, A_2^{-1}, \ldots, A_n^{-1}$.**

$$(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \cdots A_1^{-1}.$$

In either order of multiplication, associativity allows adjacent factors $A_j A_j^{-1}$ or $A_j^{-1} A_j$ to cancel successively, leaving the identity.

23. **Show that if $A \in M_n(\mathbb{K})$ is invertible, then $\overline{A}$ is also invertible and $(\overline{A})^{-1} = \overline{A^{-1}}$.**

Conjugating $AA^{-1} = A^{-1}A = I_n$ gives

$$\overline{A}\,\overline{A^{-1}} = I_n = \overline{A^{-1}}\,\overline{A}.$$

Hence $\overline{A}$ is invertible and $(\overline{A})^{-1} = \overline{A^{-1}}$.

24. **Let $A \in M_n(\mathbb{K})$ satisfy $A^3 + A = 5I_n$.**

a) Justify that A is invertible and determine A^{-1} in terms of A.

Assume $A^3 + A = 5I_n$.

Since

$$A\frac{A^2 + I_n}{5} = \frac{A^3 + A}{5} = I_n = \frac{A^2 + I_n}{5}A,$$

A is invertible and $A^{-1} = \frac{1}{5}(A^2 + I_n)$.

b) Justify that $7A$ is invertible and determine $(7A)^{-1}$ in terms of A.

As $7 \neq 0$, $7A$ is invertible and $(7A)^{-1} = \frac{1}{7}A^{-1} = \frac{1}{35}(A^2 + I_n)$.

25. **Let $A \in M_n(\mathbb{K})$ be invertible. Show that**

a) if $B \in M_{n\times m}(\mathbb{K})$ satisfies $AB = 0_{n\times m}$, then $B = 0_{n\times m}$.
$AB = 0_{n\times m}$ implies $B = A^{-1}(AB) = A^{-1}0_{n\times m} = 0_{n\times m}$.

b) if $B \in M_{m\times n}(\mathbb{K})$ satisfies $BA = 0_{m\times n}$, then $B = 0_{m\times n}$.
$BA = 0_{m\times n}$ implies $B = (BA)A^{-1} = 0_{m\times n}A^{-1} = 0_{m\times n}$.

c) if $B, C \in M_{n\times m}(\mathbb{K})$ satisfy $AB = AC$, then $B = C$.
$AB = AC$ implies $A^{-1}AB = A^{-1}AC$, hence $B = C$.

d) if $B, C \in M_{m\times n}(\mathbb{K})$ satisfy $BA = CA$, then $B = C$.
$BA = CA$ implies $BAA^{-1} = CAA^{-1}$, hence $B = C$.
All products have the dimensions specified in the question.

26. **For each of the following matrices, determine whether it is invertible and, if it is, compute its inverse.**

a) $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$
For the row reductions below, apply the stated operations to $[M \mid I]$. When the left block becomes I, the right block is M^{-1}.
$A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$ is not invertible: $F_2 \leftarrow F_2 - 4F_1$ produces a zero row, so $\mathrm{r}(A) = 1 < 2$.

b) $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$
For $B = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, use $F_2 \leftarrow F_2 - F_1$, $F_2 \leftarrow -\frac{1}{3}F_2$, $F_1 \leftarrow F_1 - F_2$. Then

$$B^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}.$$

c) $C = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
$C = \mathrm{diag}(5, -4, 7)$ is invertible. Dividing its three rows by $5, -4, 7$ gives $C^{-1} = \mathrm{diag}(\frac{1}{5}, -\frac{1}{4}, \frac{1}{7})$.

d) $D = \begin{bmatrix} 0 & 0 & 5 \\ 0 & -4 & 0 \\ 7 & 0 & 0 \end{bmatrix}$
For $D = \begin{bmatrix} 0 & 0 & 5 \\ 0 & -4 & 0 \\ 7 & 0 & 0 \end{bmatrix}$, interchange rows $1, 3$ and divide the resulting rows by $7, -4, 5$. Thus

$$D^{-1} = \begin{bmatrix} 0 & 0 & 1/7 \\ 0 & -1/4 & 0 \\ 1/5 & 0 & 0 \end{bmatrix}.$$

e) $E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is invertible. The operation $F_1 \leftarrow F_1 - 4F_3$ gives $E^{-1} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

f) $F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ satisfies $F^2 = I_3$, so $F^{-1} = F$.

g) $G = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

For $G = \begin{bmatrix} 0 & 3 & 1 \\ -1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, use

$$F_1 \leftrightarrow F_2, \quad F_1 \leftarrow -F_1, \quad F_2 \leftrightarrow F_3, \quad F_3 \leftarrow F_3 - 3F_2, \quad F_3 \leftarrow -\tfrac{1}{2}F_3,$$
$$F_2 \leftarrow F_2 - F_3, \quad F_1 \leftarrow F_1 + F_3, \quad F_1 \leftarrow F_1 + 3F_2.$$

The left block becomes I_3, and

$$G^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 1/2 & 0 & -1/2 \\ -1/2 & 0 & 3/2 \end{bmatrix}.$$

h) $H = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -2 \end{bmatrix}$

$H = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -2 \end{bmatrix}$ is not invertible. Its third column is twice its first column, so $H \begin{bmatrix} -2 & 0 & 1 \end{bmatrix}^T = 0$ with a nonzero column. If H^{-1} existed, multiplication by it would give a contradiction.

i) $I = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -3 \end{bmatrix}$

For $I = \begin{bmatrix} 2 & -1 & 4 \\ 1 & -1 & 2 \\ -1 & 0 & -3 \end{bmatrix}$, use

$$F_1 \leftarrow \tfrac{1}{2}F_1, \quad F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 + F_1, \quad F_2 \leftarrow -2F_2,$$
$$F_3 \leftarrow F_3 + \tfrac{1}{2}F_2, \quad F_3 \leftarrow -F_3, \quad F_1 \leftarrow F_1 - 2F_3, \quad F_1 \leftarrow F_1 + \tfrac{1}{2}F_2.$$

Hence I is invertible and $I^{-1} = \begin{bmatrix} 3 & -3 & 2 \\ 1 & -2 & 0 \\ -1 & 1 & -1 \end{bmatrix}$.

j) $J = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & -1 & -1 \end{bmatrix}$

$J = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & -1 & -1 \end{bmatrix}$ is not invertible: its first row is the sum of rows $2, 3$.

Equivalently, $J \begin{bmatrix} 1 & -3 & 5 \end{bmatrix}^T = 0$ with a nonzero column.

k) $K = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & -2 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ -2 & 0 & 2 & -2 \end{bmatrix}$

For $K = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 2 & -2 & 1 & 1 \\ 1 & -1 & 0 & 1 \\ -2 & 0 & 2 & -2 \end{bmatrix}$, apply

$$F_2 \leftarrow F_2 - 2F_1, \quad F_3 \leftarrow F_3 - F_1, \quad F_4 \leftarrow F_4 + 2F_1, \quad F_2 \leftrightarrow F_4,$$
$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow -F_3, \quad F_4 \leftarrow F_4 + F_3, \quad F_4 \leftarrow -\tfrac{1}{2}F_4.$$

The augmented matrix is now

$$\left[\begin{array}{cccc|cccc} 1 & -1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & -1 & -1 & 0 & 0 & -1/2 \\ 0 & 0 & 1 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1/2 & -1/2 & 1/2 & 0 \end{array}\right].$$

Complete the reduction with

$$F_1 \leftarrow F_1 - 2F_4, \quad F_2 \leftarrow F_2 + F_4, \quad F_3 \leftarrow F_3 - F_4,$$
$$F_1 \leftarrow F_1 - F_3, \quad F_2 \leftarrow F_2 + 2F_3, \quad F_1 \leftarrow F_1 + F_2.$$

Thus

$$K^{-1} = \begin{bmatrix} 0 & 1 & -2 & -1/2 \\ 1/2 & 1/2 & -5/2 & -1/2 \\ 1/2 & 1/2 & -3/2 & 0 \\ 1/2 & -1/2 & 1/2 & 0 \end{bmatrix}.$$

27. **For each $\alpha \in \mathbb{R}$, consider the matrix**

$$A_\alpha = \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 0 & \alpha + 3 & \alpha + 2 \\ 1 & \alpha & \alpha & -1 \\ -1 & 1 - \alpha & 1 & 7 \end{bmatrix}.$$

a) Determine the values of α for which A_α is invertible.

$$A_\alpha = \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 0 & \alpha+3 & \alpha+2 \\ 1 & \alpha & \alpha & -1 \\ -1 & 1-\alpha & 1 & 7 \end{bmatrix}.$$

Apply $F_3 \leftarrow F_3 - F_1$, $F_4 \leftarrow F_4 + F_1$, $F_2 \leftrightarrow F_4$, $F_3 \leftrightarrow F_4$. Then

$$A_\alpha \sim \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ 0 & 1 & \alpha+1 & \alpha+7 \\ 0 & 0 & \alpha+3 & \alpha+2 \\ 0 & 0 & 0 & -\alpha-1 \end{bmatrix}.$$

For $\alpha \notin \{-3,-1\}$, there are four pivots. For $\alpha = -1$ the last row is zero. For $\alpha = -3$, the last two rows have nonzero entries only in column 4 and are dependent. Therefore

$$A_\alpha \text{ is invertible} \iff \alpha \in \mathbb{R} \setminus \{-3,-1\}.$$

b) Compute the inverse of A_0.

For $\alpha = 0$, apply the same operations to $[A_0 \mid I_4]$:

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 7 & 1 & 0 & 0 & 1 \\ 0 & 0 & 3 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 \end{array}\right].$$

Now use $F_4 \leftarrow -F_4$, $F_3 \leftarrow F_3 - 2F_4$, $F_3 \leftarrow \frac{1}{3}F_3$, $F_2 \leftarrow F_2 - 7F_4$, $F_2 \leftarrow F_2 - F_3$. The left block is I_4, giving

$$A_0^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -16/3 & -1/3 & 19/3 & 1 \\ -2/3 & 1/3 & 2/3 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}.$$

28. **Compute the inverse of the matrix**

$$A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2i & -1 \\ 2 & -4 & i \end{bmatrix}.$$

For $A = \begin{bmatrix} 1 & 0 & i \\ 0 & 2i & -1 \\ 2 & -4 & i \end{bmatrix}$, apply to $[A \mid I_3]$ the operations $F_3 \leftarrow F_3 - 2F_1$, $F_2 \leftarrow -\frac{i}{2}F_2$, $F_3 \leftarrow F_3 + 4F_2$, $F_3 \leftarrow -iF_3$. This gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & i & 1 & 0 & 0 \\ 0 & 1 & i/2 & 0 & -i/2 & 0 \\ 0 & 0 & 1 & 2i & -2 & -i \end{array}\right].$$

Finally, $F_1 \leftarrow F_1 - iF_3$ and $F_2 \leftarrow F_2 - \frac{i}{2}F_3$ give

$$A^{-1} = \begin{bmatrix} 3 & 2i & -1 \\ 1 & i/2 & -1/2 \\ 2i & -2 & -i \end{bmatrix}.$$

29. **Let**

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix}.$$

a) Compute the inverse of B.

Here

$$A = \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -2 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix}.$$

On $[B \mid I_3]$, use

$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow F_3 - F_2, \quad F_3 \leftarrow \tfrac{2}{3}F_3, \quad F_2 \leftarrow F_2 + \tfrac{1}{2}F_3, \quad F_1 \leftarrow F_1 - 2F_2.$$

Then B is invertible and

$$B^{-1} = \begin{bmatrix} 1 & 2/3 & -2/3 \\ 0 & -1/3 & 1/3 \\ 0 & 1/3 & 2/3 \end{bmatrix}.$$

b) Solve $AX = 2X + BC + C$ for $X \in M_{3\times 2}(\mathbb{R})$.

Since $A - 2I_3 = B$, the equation is equivalent to $BX = BC + C$. Multiplying on the left by B^{-1} gives the unique solution

$$X = C + B^{-1}C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 5/3 & 4/3 \\ -1/3 & -2/3 \\ -2/3 & 2/3 \end{bmatrix} = \begin{bmatrix} 8/3 & 4/3 \\ -1/3 & 4/3 \\ -5/3 & 2/3 \end{bmatrix}.$$

Substitution gives $BX = BC + C$, hence the original equation.

30. **Let $\boldsymbol{A} \in \boldsymbol{M}_{m\times n}(\mathbb{K})$ and let $\boldsymbol{P} \in \boldsymbol{M}_m(\mathbb{K})$ be invertible. Show that $\mathrm{r}(\boldsymbol{PA}) = \mathrm{r}(\boldsymbol{A})$.**

Since P is invertible, row reduction takes it to I_m: $E_s \cdots E_1 P = I_m$, where the E_j are elementary matrices. Thus $P = E_1^{-1} \cdots E_s^{-1}$. Each inverse is elementary, so PA is obtained from A by elementary row operations. Such operations preserve rank; therefore $\mathrm{r}(PA) = \mathrm{r}(A)$.

31. **Determine whether each of the following equations is linear in the unknowns $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$.**

a) $x + \pi y + \sqrt[3]{2}\, z = e$.

An equation is linear in x, y, z if it has the form $ax + by + cz = d$ with fixed coefficients and no products or nonlinear functions of the unknowns.

$x + \pi y + \sqrt[3]{2}\, z = e$: linear; all coefficients are constants.

b) $x^2 + xy + z = 2$.

$x^2 + xy + z = 2$: not linear, because of x^2 and xy.

c) $x + \sin y + 2z = 0$.

$x + \sin y + 2z = 0$: not linear, because of $\sin y$.

d) $4x + (\sin \frac{\pi}{10})^2 y + 23z = \cos \frac{\pi}{5}$.

$4x + (\sin \frac{\pi}{10})^2 y + 23z = \cos \frac{\pi}{5}$: linear; the trigonometric expressions and 23 are constants.

e) $(-x + 8y - 2z)^2 = 1$.

$(-x + 8y - 2z)^2 = 1$: not linear. It is equivalent to the union of the two equations $-x + 8y - 2z = 1$ and $-x + 8y - 2z = -1$, not to a single linear equation.

f) $ax + 6y - 3z = 9$, with $a \in \mathbb{R}$ fixed.

$ax + 6y - 3z = 9$, with fixed $a \in \mathbb{R}$: linear, including when $a = 0$.

32. **Solve the following systems by Gauss–Jordan elimination. Then give three particular solutions and the general solution of the associated homogeneous system.**

a)

$$\begin{cases} x_1 - 3x_2 + 2x_3 + 2x_4 + 7x_5 = 8, \\ x_1 - 3x_2 + 2x_3 + 3x_4 + 11x_5 = 11, \\ 3x_1 - 9x_2 + 6x_3 - 3x_4 - 15x_5 = -3, \\ -2x_1 + 6x_2 - 4x_3 + 2x_4 + 10x_5 = 2. \end{cases}$$

The unknowns are x_1, x_2, x_3, x_4, x_5.

The augmented matrix is

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 2 & 2 & 7 & 8 \\ 1 & -3 & 2 & 3 & 11 & 11 \\ 3 & -9 & 6 & -3 & -15 & -3 \\ -2 & 6 & -4 & 2 & 10 & 2 \end{array}\right].$$

Apply

$$F_2 \leftarrow F_2 - F_1, \quad F_3 \leftarrow F_3 - 3F_1, \quad F_4 \leftarrow F_4 + 2F_1,$$
$$F_3 \leftarrow F_3 + 9F_2, \quad F_4 \leftarrow F_4 - 6F_2, \quad F_1 \leftarrow F_1 - 2F_2.$$

The reduced augmented matrix is

$$\left[\begin{array}{rrrrr|r} 1 & -3 & 2 & 0 & -1 & 2 \\ 0 & 0 & 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Put $x_2 = s$, $x_3 = t$, $x_5 = u$. The general solution is

$$(x_1, x_2, x_3, x_4, x_5) = (2, 0, 0, 3, 0) + s(3, 1, 0, 0, 0) + t(-2, 0, 1, 0, 0) + u(1, 0, 0, -4, 1).$$

There are three free unknowns. Three particular solutions are $(2, 0, 0, 3, 0)$, $(5, 1, 0, 3, 0)$ and $(0, 0, 1, 3, 0)$. The general solution of the associated homogeneous system is

$$s(3, 1, 0, 0, 0) + t(-2, 0, 1, 0, 0) + u(1, 0, 0, -4, 1), \qquad s, t, u \in \mathbb{R}.$$

b)

$$\begin{cases} 3x_1 + 7x_3 + 5x_4 = 21, \\ x_1 + 3x_3 + 3x_4 = 9, \\ 2x_1 + 5x_3 + 4x_4 = 15, \\ x_1 + 2x_3 + x_4 = 6. \end{cases}$$

The unknowns are x_1, x_2, x_3, x_4.
The augmented matrix is

$$\left[\begin{array}{cccc|c} 3 & 0 & 7 & 5 & 21 \\ 1 & 0 & 3 & 3 & 9 \\ 2 & 0 & 5 & 4 & 15 \\ 1 & 0 & 2 & 1 & 6 \end{array}\right].$$

Apply

$$F_1 \leftrightarrow F_2, \quad F_2 \leftarrow F_2 - 3F_1, \quad F_3 \leftarrow F_3 - 2F_1, \quad F_4 \leftarrow F_4 - F_1,$$
$$F_2 \leftarrow -\tfrac{1}{2}F_2, \quad F_3 \leftarrow F_3 + F_2, \quad F_4 \leftarrow F_4 + F_2, \quad F_1 \leftarrow F_1 - 3F_2.$$

This gives

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Putting $x_4 = s$, $x_2 = t$, the general solution is

$$(x_1, x_2, x_3, x_4) = (0, 0, 3, 0) + s(3, 0, -2, 1) + t(0, 1, 0, 0).$$

Three particular solutions are $(0, 0, 3, 0)$, $(0, 1, 3, 0)$ and $(3, 0, 1, 1)$. The associated homogeneous system has general solution $s(3, 0, -2, 1) + t(0, 1, 0, 0)$, with $s, t \in \mathbb{R}$.

33. **State, with justification, whether each of the following assertions is true or false for any linear system (S) with m equations and n unknowns.**

a) If $m > n$, then (S) is inconsistent.
False. The system $x = 1$, $2x = 2$ has $m = 2 > 1 = n$ and a unique solution.

b) If $m < n$, then (S) is inconsistent.

False. The equation $x + y = 0$ has $m = 1 < 2 = n$ and has solutions.

c) If $m < n$, then (S) is either inconsistent or has infinitely many solutions.

True. If the system is consistent, its number of pivots is at most $m < n$, so at least one unknown is free. Thus it has infinitely many solutions. Otherwise it is inconsistent.

d) If (S) has a unique solution, then $m = n$.

False. The counterexample in (a) has a unique solution with $m \neq n$.

e) If $m = n$, then (S) has a unique solution.

False. The system $x + y = 0$, $2x + 2y = 0$ has $m = n = 2$ and infinitely many solutions. Also, $x + y = 0$, $x + y = 1$ is inconsistent with $m = n = 2$.

34. **Discuss the following systems in the unknowns x, y, z in terms of the parameter a. When there are infinitely many solutions, state the degree of indeterminacy.**

a)

$$\begin{cases} x + y - az = 0, \\ ax + y - z = 1 - a, \\ x + ay - z = 1 - a. \end{cases}$$

The system is

$$x + y - az = 0, \qquad ax + y - z = 1 - a, \qquad x + ay - z = 1 - a.$$

Subtract the first equation from the second and third:

$$(a - 1)(x + z) = 1 - a, \qquad (a - 1)(y + z) = 1 - a.$$

For $a = 1$, all three original equations reduce to $x + y - z = 0$. The general solution is $(x, y, z) = (s, t, s + t)$; the degree of indeterminacy is 2.

For $a \neq 1$, $x + z = y + z = -1$. Substitution in the first equation gives $(a + 2)z = -2$. If $a = -2$, this is impossible. If $a \notin \{-2, 1\}$, the unique solution is

$$(x, y, z) = \left(-\frac{a}{a+2}, -\frac{a}{a+2}, -\frac{2}{a+2} \right).$$

b)

$$\begin{cases} x + y - z = 0, \\ ay + (2a - 1)z = a, \\ ax + 2ay = 0, \\ ax + ay + (1 - 2a)z = a - 1. \end{cases}$$

The four equations are

$$x + y - z = 0, \qquad ay + (2a - 1)z = a,$$
$$ax + 2ay = 0, \quad ax + ay + (1 - 2a)z = a - 1.$$

The left side of equation 2 plus equation 4 minus equation 3 is zero; the same combination on the right is $2a - 1$. Consistency therefore requires $a = \frac{1}{2}$. At this value, equation 2 gives $y = 1$, equation 3 gives $x = -2$, and equation 1 gives $z = -1$. These values satisfy equation 4. Thus the system is uniquely solvable for $a = \frac{1}{2}$, with solution $(-2, 1, -1)$, and is inconsistent for every other a. There is no indeterminate case.